



Fig. 4: Circuit printed using Argentum (Image courtesy: Cartesian Co.)

ible and rigid electronic circuits on any material of your choice (polyimide, linen paper, fibre glass, synthetic paper or fabric), and it is as easy as printing a photo.

The method. Argentum comes with two inkjet cartridges similar to the ones in a desktop printer, and each cartridge is filled with a different chemical. First, it prints in one chemical, followed by the other chemical on top of the existing one. When these two chemicals mix, a reaction occurs and produces silver nanoparticles, leaving a silver image on the substrate.

Resolution and speed. The device can natively print images at 300 dots-per-inch (DPI), which can be enhanced to 600DPI using the accompanying software. The circuit can be created using any electronics circuit design tool including Microsoft Paint, and can be easily exported to the printer software to print the circuit on the selected substrate. The printed circuit board (PCB) does not have to be etched either. This reduces the overall time to design, create and test the circuit.

Small form factor, big print areas. The size of Argentum is about 43cm × 32cm × 17cm and it weighs around 6kg, which is close to a normal desktop inkjet printer. The user can print circuits with a print area of 17.5cm × 8cm, which is well above the size of what one can design in the free version of EAGLE software.

Soldering. Circuits printed using Argentum can be reflow-soldered with the conventional hot plate, reflow oven and hot air-gun, using leaded solder paste. It can also be hand-soldered, though it needs to be done quickly and adeptly to avoid damaging the silver traces.

Commercialisation on demand

According to a report by IDTechX titled 'Printed, Organic & Flexible Electronics Forecasts, Players & Opportunities 2015-2025', the market for printed, flexible and organic electronics is set to increase from US\$ 29.80 billion in 2015 to US\$ 73.69 billion in 2025.

Currently, we are in the early days of electronics printing, but with this increased market pull, we can be optimistic about more researchers and companies coming up with methods to print electronics circuits in mass quantities. ●

www.indiaelectronicsweek.com

EFY GROUP
INDIA

India's first event to focus on the electronics that powers LED Lighting



"Make
in India"
Offer

Special package for those who manufacture
LEDs and LED Lighting products in India!

To increase your sales by exhibiting in this event, Call:
Jyotsna (011-40596605) or write to us at: efyexpo@efy.in

Co-located at

INDIA
ELECTRONICS
WEEK

January 11th-13th, 2016
Bengaluru